

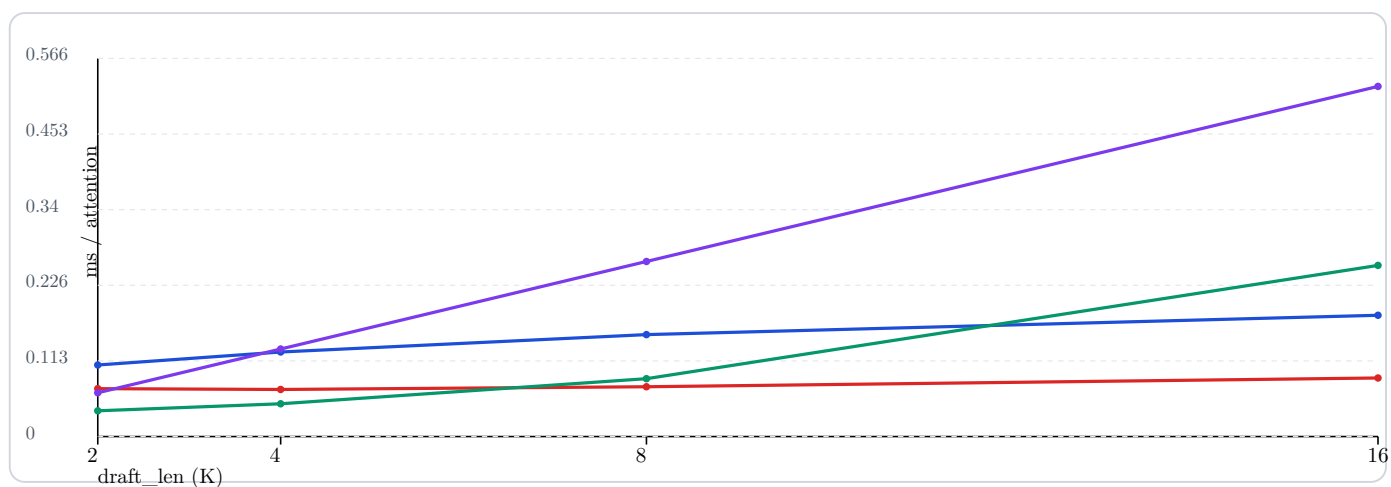
A40 Decode Attention Speed (Final)

Runs: A40 spot, bf16, implementation=cudnn, prefix_len=1024, iterations=100, repeats=5, batch_size=1.

Pallas series is intentionally omitted from these plots to keep the non-pallas curves readable on a single scale. AR baseline is measured at batch=1, query_len=1 and scaled by K for each draft length point.

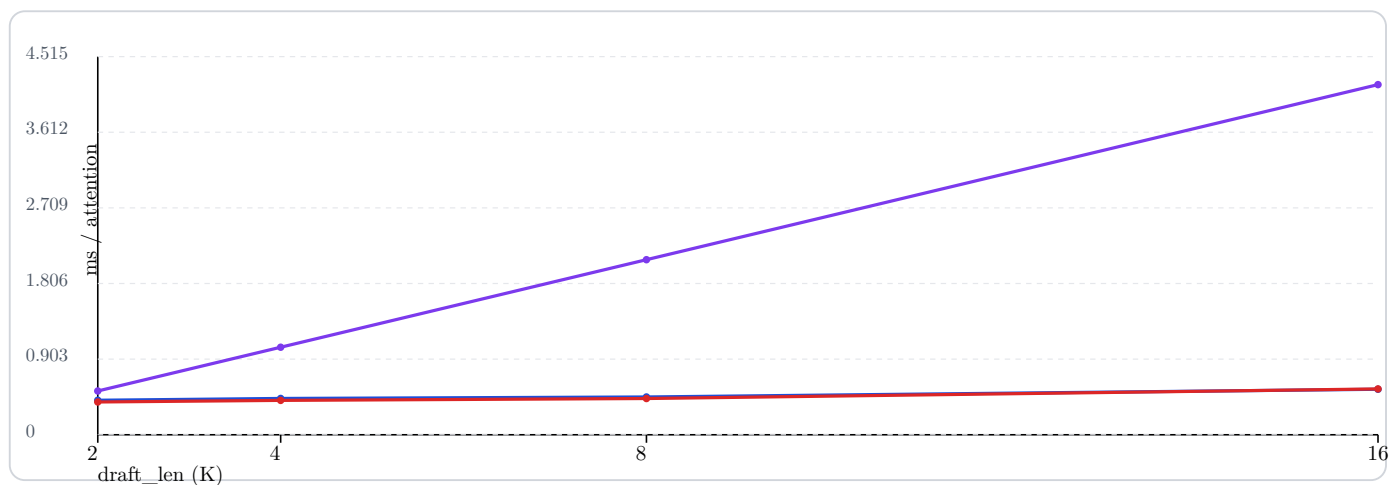
1B-style MHA (32/32, head_dim 64) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



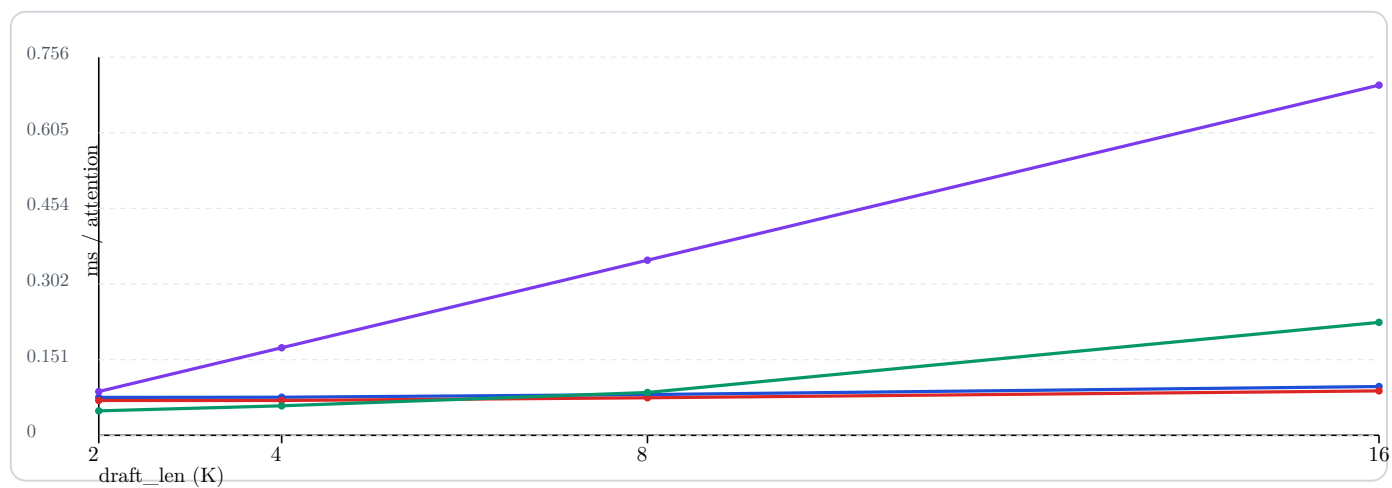
1B-style MHA (32/32, head_dim 64) - Native decode

- native structured
- native dense
- AR baseline (K x len1)



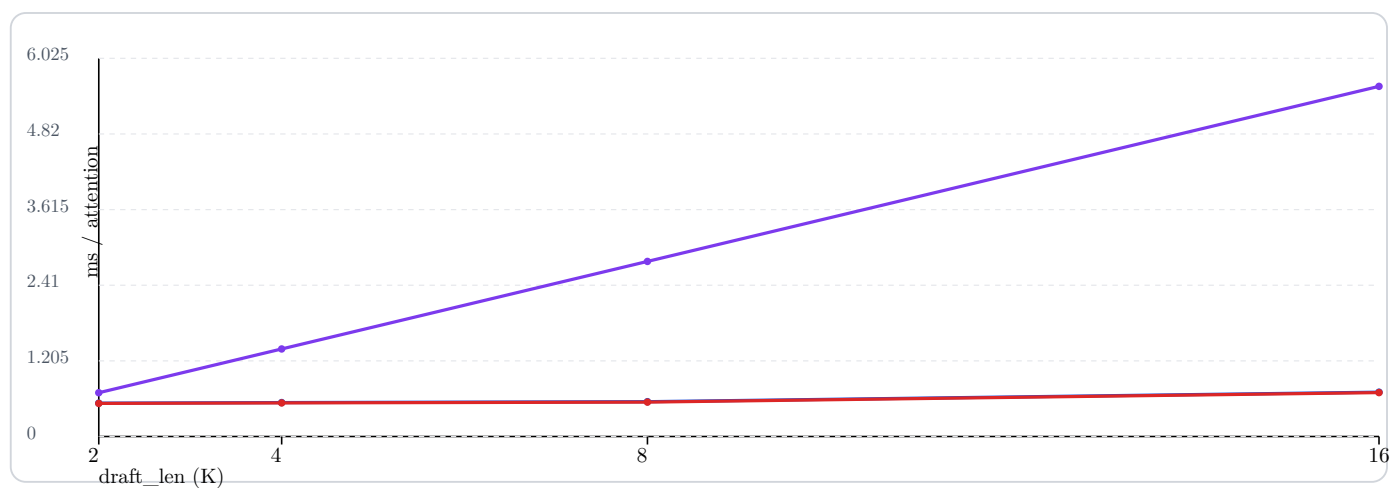
3B-style GQA (24/8, head_dim 128) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



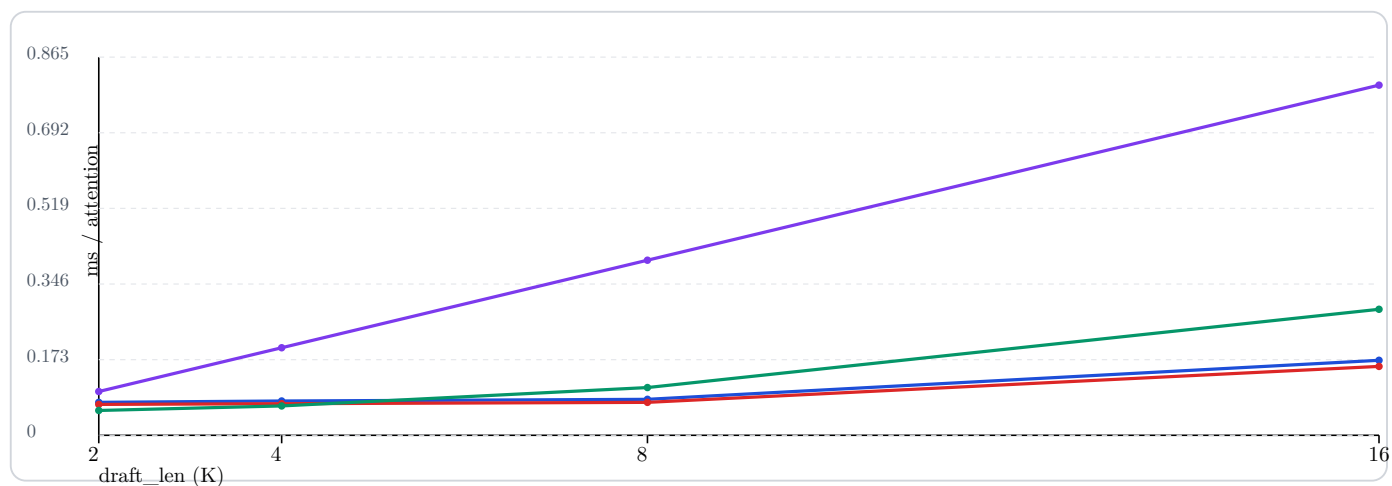
3B-style GQA (24/8, head_dim 128) - Native decode

- native structured
- native dense
- AR baseline (K x len1)



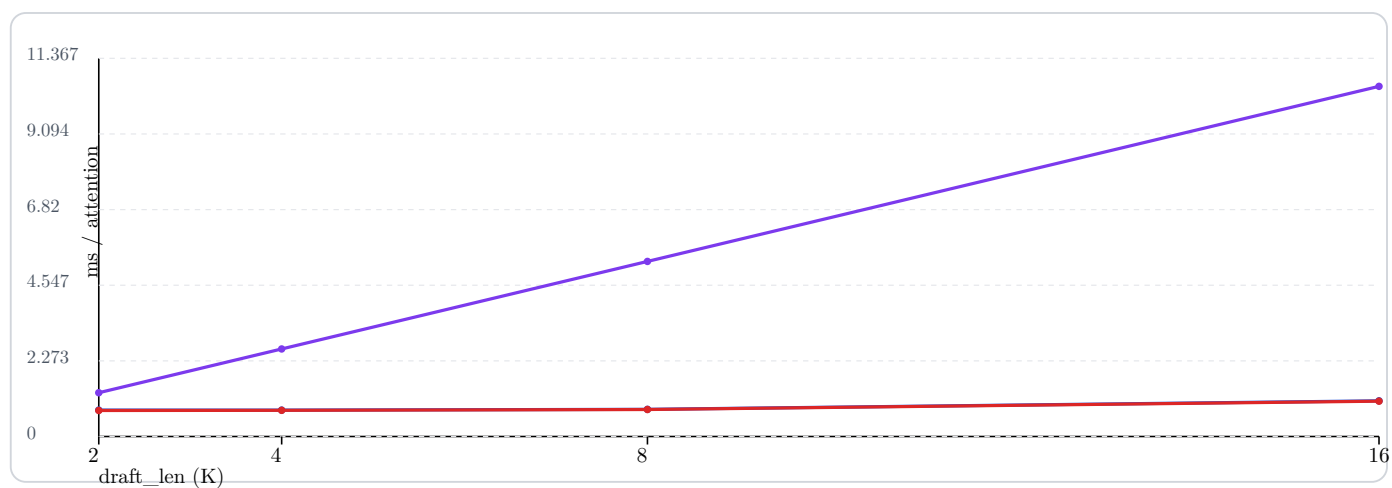
7B-style GQA (32/8, head_dim 128) - Kernel terms

- kernel structured
- kernel dense
- kernel dense+jax-zero
- AR baseline (K x len1)



7B-style GQA (32/8, head_dim 128) - Native decode

- native structured
- native dense
- AR baseline (K x len1)

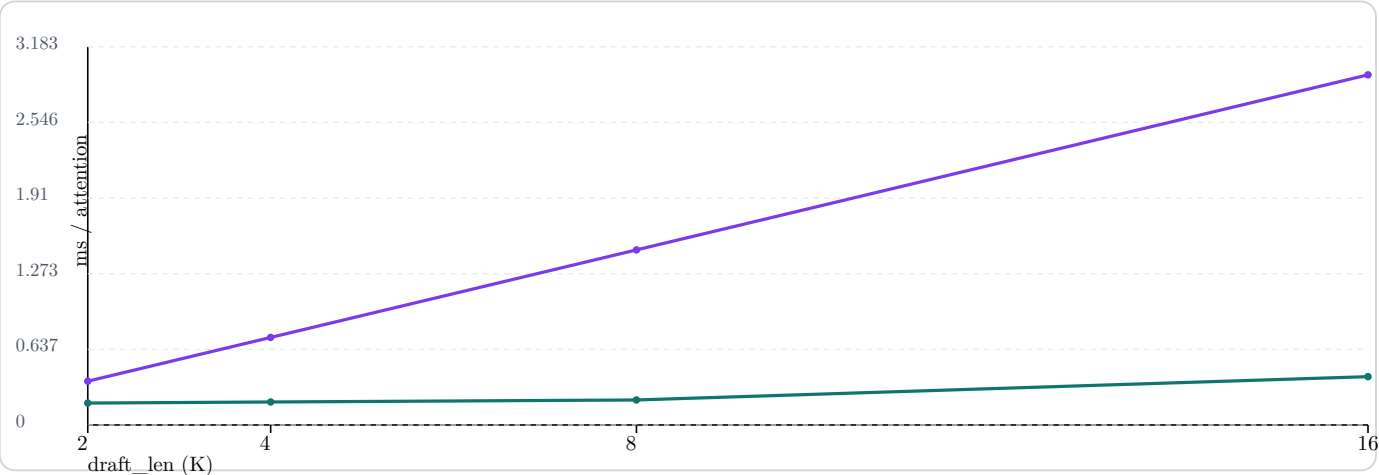


A40 MLP Throughput — AR vs TiDAR Tokens

This section compares MLP-only cost using SwiGLU projections. AR baseline is measured on `len=1` and scaled by `K`, while TiDAR runs the full $L = K + K^2$ token block in one pass.

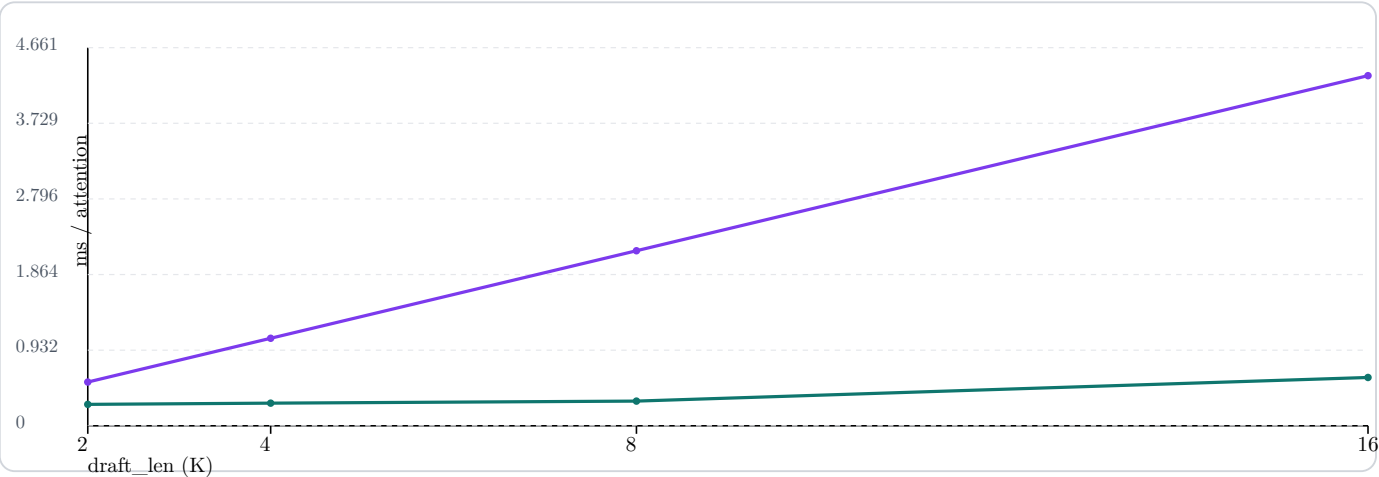
1B-style MHA (`d_model=2048`, `d_ff=8192`) - MLP only

- TiDAR MLP at $L=K+K^2$
- AR MLP baseline ($K \times len1$)



3B-style GQA (`d_model=3072`, `d_ff=8192`) - MLP only

- TiDAR MLP at $L=K+K^2$
- AR MLP baseline ($K \times len1$)



7B-style GQA (d_model=4096, d_ff=11008) - MLP only

TiDAR MLP at $L=K+K^2$

AR MLP baseline ($K \times \text{len1}$)

